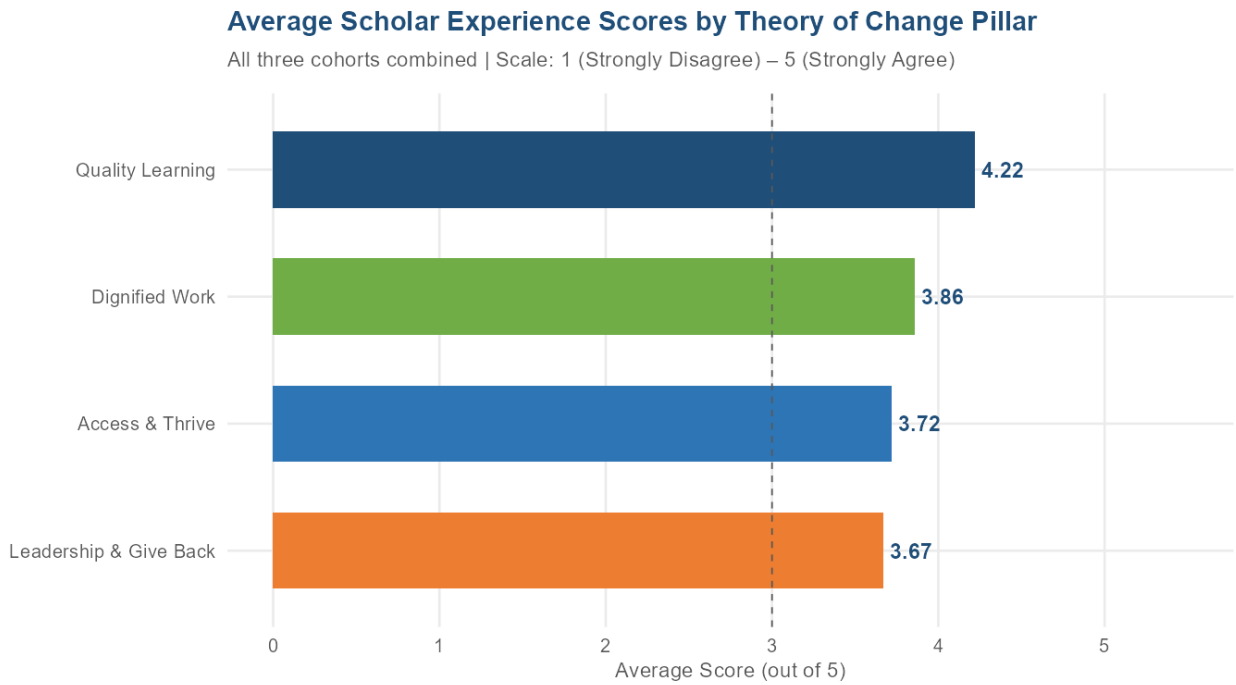
Monitoring Scholar Success: A Data-Driven Evaluation of an African Youth Scholarship Program

Disclaimer: The TechRise Scholar Program is entirely fictional and created for portfolio demonstration purposes only. All data used in this project is simulated and does not represent any real organisation, program or individual. The analysis, results and recommendations are based solely on this simulated data.

Results & Analysis

RQ1: How are scholars experiencing the program?



The short answer: Scholars are having a generally positive experience across all four pillars, every pillar scores above the midpoint of 3.0. But the scores are not equal and the gaps matter.

What this means in plain language:

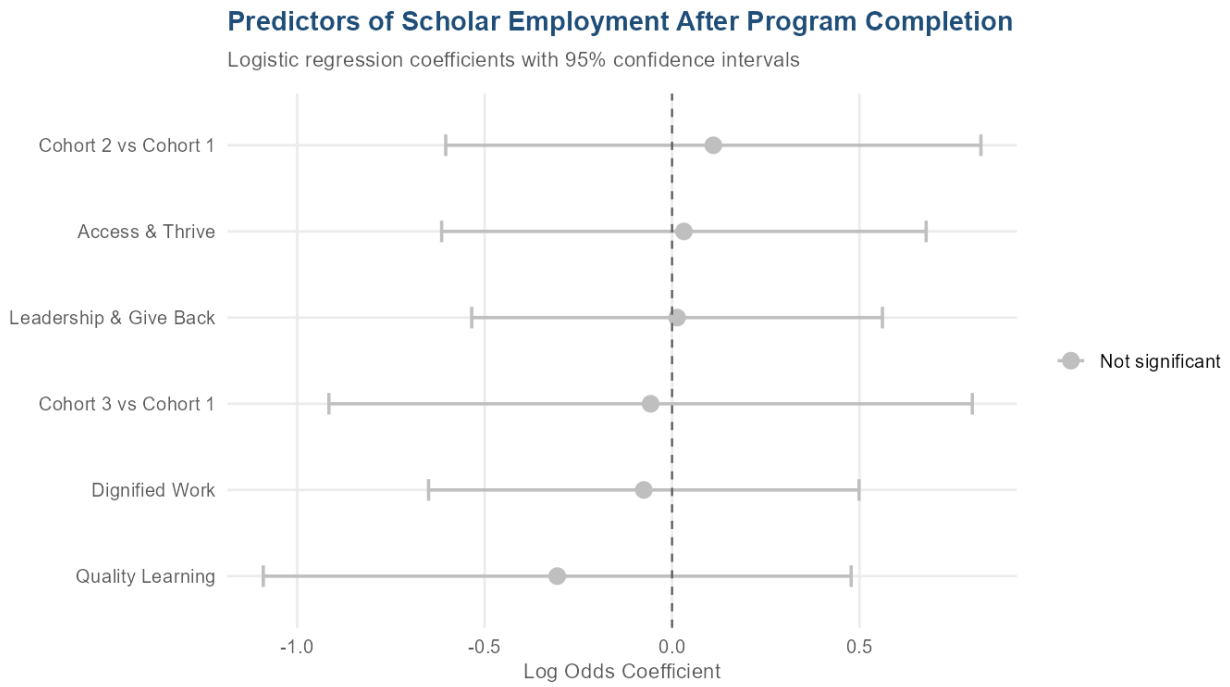
Quality Learning: Scholars feel the most positive about the quality of training they are receiving , they find the content relevant, the instruction good and they feel they are gaining practical skills. This is the program's biggest win and a strong foundation to build on.

Dignified Work: They feel reasonably confident about their work prospects after graduation, though this pillar has room to grow, particularly around career planning support.

Access & Thrive : The score indicates that most scholars feel safe and supported but the score is lower than Quality Learning, which is a signal that not all scholars are accessing the program equally. This is explored further in RQ4.

Leadership & Give Back is the weakest pillar, scholars feel the least equipped for community leadership and least engaged with Give Back activities. This is the program's biggest growth opportunity and the good news is that it is also showing the steepest improvement across cohorts (see RQ3).

RQ2 : What predicts whether a scholar completes the program and gets a job?



The short answer: None of the individual factors tested were statistically significant predictors of employment in this data, which itself is a finding worth taking seriously.

What the analysis did:

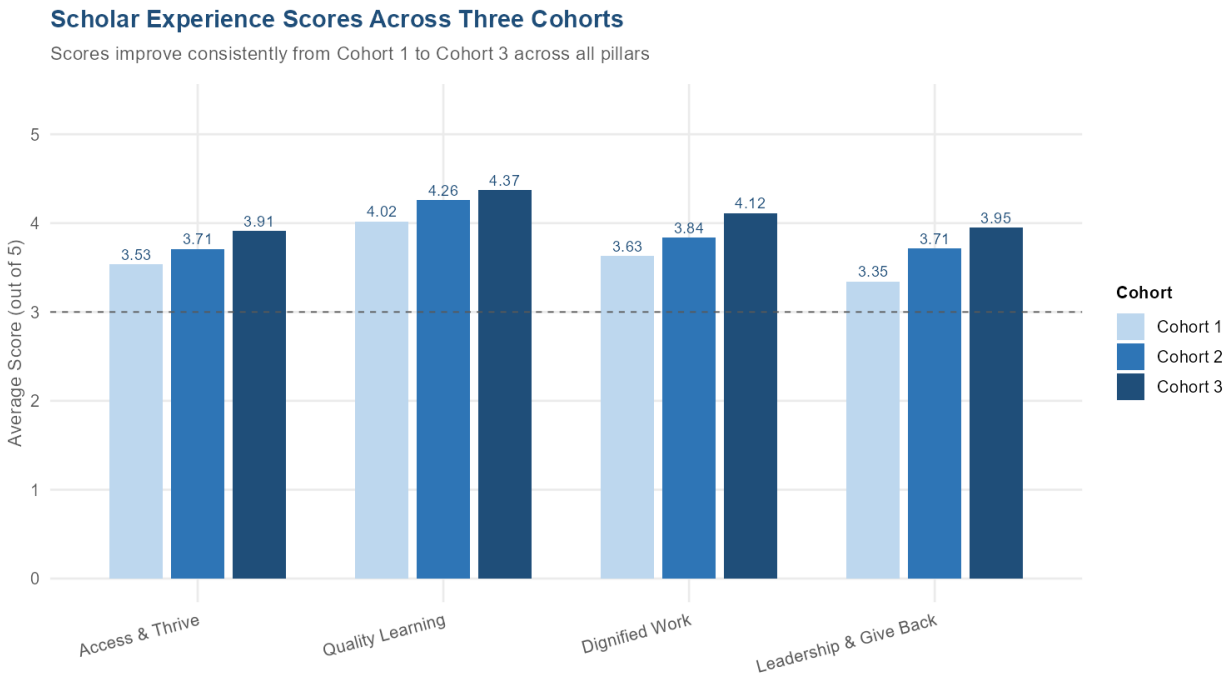
A logistic regression was run with employment as the outcome. The predictors tested were all four pillar scores, cohort and refugee status.

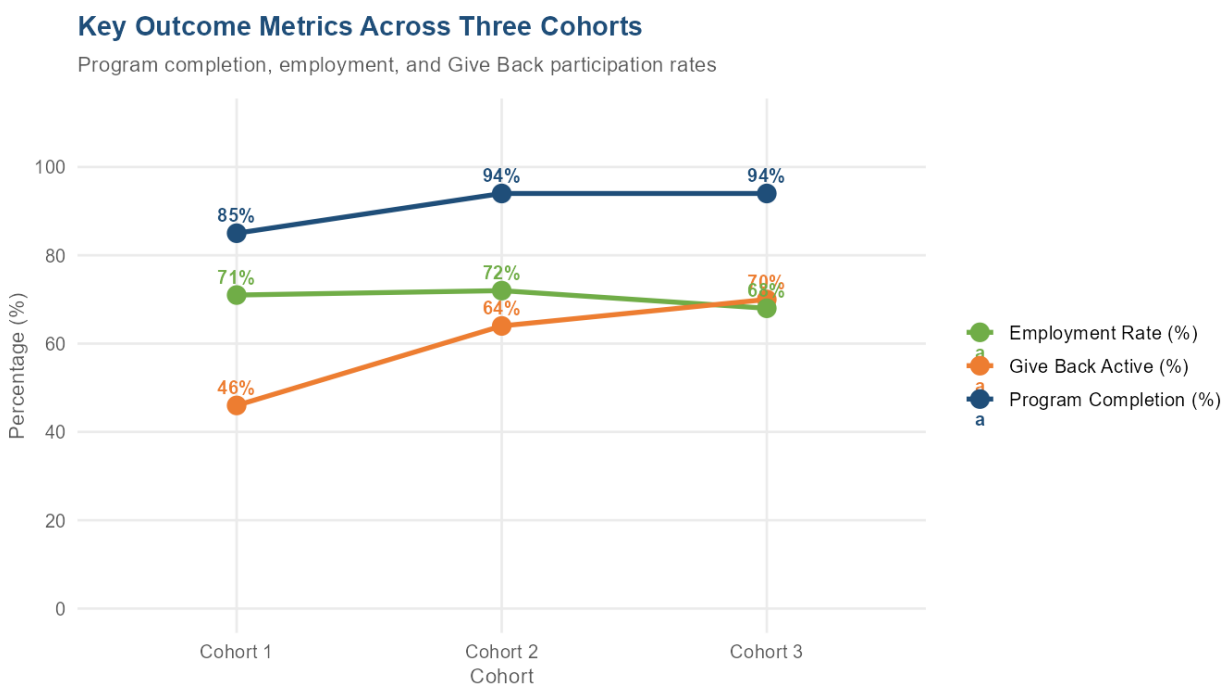
What this means in plain language:

The fact that no single factor emerged as a statistically significant predictor suggests that employment outcomes in this program are likely multi-factorial, it is not just one thing that gets a scholar a job. It is probably a combination of training quality, career support, personal networks and local labor market conditions that together determine employment outcomes.

This is a common result in youth employment programs and has a practical implication, the program cannot rely on any one pillar to drive employment. Improving outcomes across all four pillars simultaneously is likely more effective than focusing narrowly on any single one.

RQ3: Is the program getting better over time?





The short answer: Yes, the improvement is consistent and meaningful.

Employment rate in Cohort 3 is slightly lower than Cohort 2. This is expected, Cohort 3 graduated more recently, so fewer scholars have had time to secure employment. This is a recency effect, not a decline in program quality.

What this means in plain language:

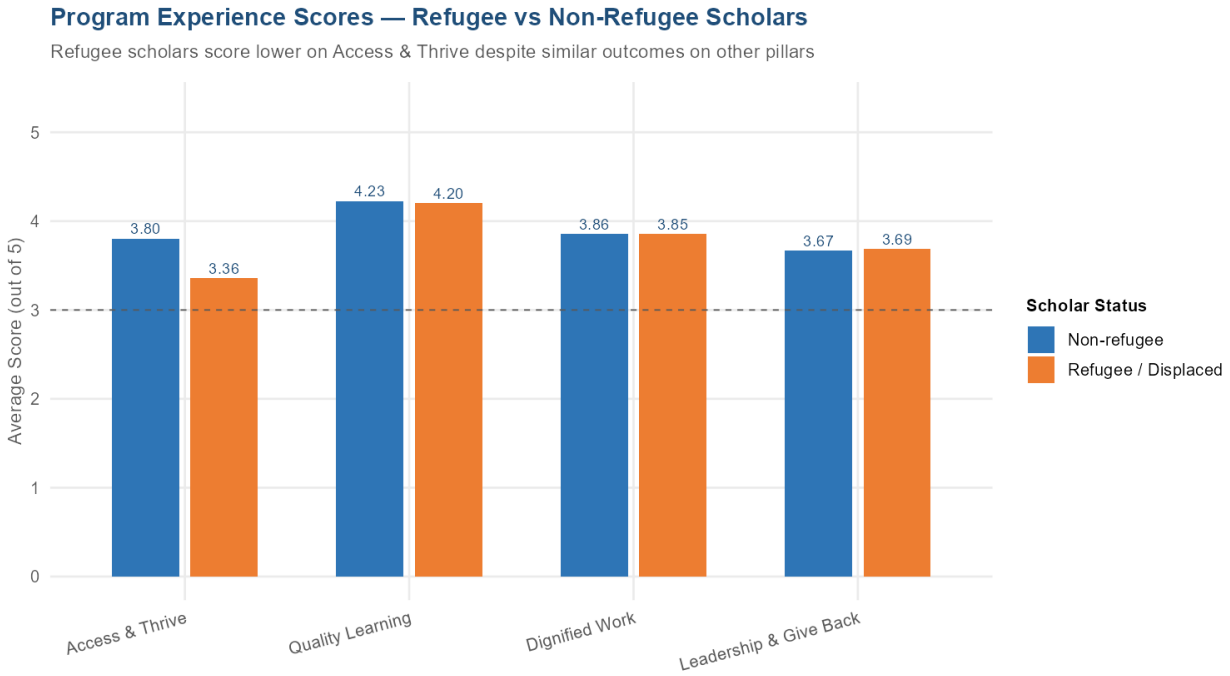
The program is learning and improving with every cohort. This is the most encouraging finding in the

entire data. The Leadership & Give Back pillar showed the biggest growth, a 0.60-point improvement over three cohorts, suggesting that intentional program changes are strengthening scholars' sense of community responsibility and leadership identity.

The completion rate increased from 85% to 94% between Cohorts 1 and 2 is particularly significant. A 9-percentage-point improvement in one cohort indicates a meaningful operational improvement, possibly better on-boarding, stronger support structures or more targeted early intervention for at-risk scholars.

The ANOVA confirmed these differences are statistically significant ($p < 0.05$ for all pillars), meaning the improvement is not due to chance.

RQ4: Are refugee scholars experiencing the program differently?



The short answer: Yes but only on one specific pillar and the result is both important and actionable.

What the data shows:

Pillar	Non-Refugee	Refugee/Displaced	Difference	Significant?
Access & Thrive	3.80	3.36	-0.44	Yes ($p < 0.001$)
Quality Learning	4.23	4.20	-0.03	No
Dignified Work	3.86	3.85	-0.01	No
Leadership & Give Back	3.67	3.69	+0.02	No

What this means in plain language:

Refugee and displaced scholars are experiencing the training itself just as well as their non-refugee peers, their Quality Learning, Dignified Work and Leadership scores are almost identical. This is a strong signal that the curriculum and instruction are equitable.

However, they are scoring significantly lower on Access & Thrive, meaning they are facing more difficulty with device and internet access, financial stability and sense of belonging. This is not a curriculum problem, it is a wraparound support problem.

The most powerful insight is that refugee scholars are completing the program and participating in Give Back at similar rates despite these access challenges. This means they are resilient and highly motivated but they are doing it the hard way, without the same foundation support as their peers.

Recommendations

Based on the findings above, here are five evidence-based recommendations for the TechRise Scholar Program:

1. Introduce a Refugee & Displaced Scholar Support Package

Based on: RQ4: Access & Thrive gap (3.36 vs 3.80, $p < 0.001$)

Create a dedicated support package for refugee and displaced scholars that includes:

- A device lending program or data bundle for scholars without reliable internet

- A financial hardship fund for urgent needs (transport, food, documentation)

- A buddy system pairing refugee scholars with non-refugee peers in the first two weeks

This directly addresses the statistically significant gap without changing anything about the curriculum because the curriculum is already working for these scholars.

2. Invest More in Leadership & Give Back Programs

Based on: RQ1: Leadership & Give Back is the lowest-scoring pillar (3.67)

Despite showing the greatest improvement across cohorts, this pillar is still the weakest. Practical steps include:

- Introducing a structured Give Back session in the final month of the program, where scholars plan a community activity.

- Inviting alumni mentors to share their Give Back stories at the start of the program, not the end.

- Building a scholar alumni network that makes it easy for graduates to stay connected and mentor future cohorts.

3. Strengthen Career-to-Employment Transition Support

Based on: RQ2: Employment outcomes are multi-factorial, Dignified Work scores vary by cohort

Employment is the outcome the program is ultimately measured on and the data suggest it cannot be driven by any single program element. Recommended actions:

- Add a dedicated career coaching module in the final 4 weeks

- Build partnerships with employers and freelance platforms for direct job placement referrals

- Track employment at 3 months and 6 months post-graduation, not just at program end to get a more accurate picture of employment outcomes

4. Formalise Early Warning Indicators

Based on: RQ3: Completion rate increased from 85% to 94% between Cohorts 1 and 2

The program has clearly learned something between Cohorts 1 and 2 that dramatically improved completion. Formalize whatever worked:

- Identify scholars with low Access & Thrive scores at the midpoint of the program (Week 8) and trigger targeted support.

- Create a simple scholar risk score, a composite of access, financial and belonging indicators that flags scholars who need a welfare check.

- Document the intervention so it can be replicated and improved in future cohorts.

5. Administer a Midpoint Survey

Based on: The current endline-only design means problems are only identified after the program ends

This project's survey is endline-only, which means all results describe what already happened, rather than enabling real-time intervention. The midpoint survey at Week 8 would:

- Identify scholars at risk of dropping out before they do.

- Give program staff time to act on access and well-being issues.

- Enable a before/after comparison that strengthens the program's evidence base for donor reporting.

Limitations

- **Simulated data:** All findings are based on synthetically generated data. Real-world program data would produce different results and may reveal patterns not captured here.
- **Short survey instrument:** With only 2-3 Likert items per pillar, internal reliability (Cronbach's alpha) is lower than ideal. A full survey with 5-6 items per pillar would strengthen the validity of pillar scores. This is acknowledged in the methodology and would be addressed in a full-scale version of the survey with more items per pillar.
- **Endline only:** Without midpoint data, it is impossible to measure within-program growth for individual scholars. A longitudinal design would significantly strengthen the evaluation.
- **Self-reported data:** All responses are self-reported and subject to social desirability bias, scholars may rate their experience more positively than their actual experience to avoid negative judgment.
- **Employment recency effect:** Employment rates for Cohort 3 are slightly lower than Cohort 2, likely because recent graduates have had less time to secure employment. A 3-month and 6-month follow-up survey would address this.

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